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The relationship between quantified EEG and skin conductance level

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Abstract

Electroencephalographic measures (EEG) and skin conductance level (SCL) respectively reflect cerebral cortical activity and sympathetic autonomic activity. Such central and autonomic activities associated with arousal generally have been studied separately, despite their potential to reflect complementary dimensions of reticular-thalamo-hypothalamo-cortical activating networks. In this study, we examined the relationship between cortical (19 EEG sites) and autonomic (SCL) activities recorded simultaneously in 10 normal adults. Two second pre-stimulus EEGs and SCLs were assessed from an habituation paradigm which presented 22 trains of 7 tones in an 'ignore' condition. The mean SCLs of the epochs across subjects showed an initial rise (sensitization) followed by an exponential decline (habituation). Although EEG associated with the tones did not demonstrate such a distinct profile, EEG total power and band powers (beta, alpha and theta) associated with the trains showed a systematic increasing response profile. In the group data the mean SCLs within trains showed a significant correlation with alpha and beta band powers. Finer EEG band analyses indicated that $\beta 3$ at Fz and $\alpha 2$ at Cz showed the strongest separate linear correlations with SCL. $\beta 3$ and $\alpha 1$ at Fz were found to jointly covary with SCL. The findings indicate a substantive relationship between measures of cerebral function and autonomic arousal.

Keywords: Skin conductance level; Theta activity; Alpha activity; Beta activity; EEG band power; Autonomic nervous system; Central nervous system; Arousal

1. Introduction

Skin conductance level (SCL) reflects an output from the autonomic nervous system (Wallis, 1981; Hoeldtke et al., 1992) and is a peripheral measure of physiological arousal (Raine et al., 1990). The EEG, reflecting activity in the cerebral cortex, changes in its frequency distribution with the level of arousal (Daniel, 1966; Shaw and McLachlan, 1968; Santamaria and Chiappa, 1987; Makeig and Inlow, 1993). Underlying these measures are shared reticulo-thalamo-hypothalamo-cortical networks, and thus a functional link between the two measures might be predicted. However, cerebral and autonomic activities associated with arousal generally have been studied separately (Daniel, 1966; Gruzelier and Venables, 1973; Matousek and Petersén, 1983).

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Previous studies that have examined concurrent SCL and EEG activity have been focussed on alpha activity (Stennett, 1956; Davies and Krkovic, 1964; Shaw and McLachlan, 1968) and have been limited to a small number of EEG recording channels (Burch and Greiner, 1960; Volavka et al., 1967; Siddle et al., 1974; Rockstroh et al., 1988; Raine et al., 1990; Weitkunat et al., 1990; Gruzelier et al., 1993). A few studies have used a reasonable number of head sites and bilateral conductance recordings (Rippon, 1990). Inconsistent findings were found in reports on hemispheric asymmetry using bilateral electrodermal activity after a review of 51 papers (published between 1937 and 1982) on bilateral cortical and electrodermal activities (Hugdahl, 1984). Rippon, in her recent review, painted a disappointing picture of the usefulness of electrodermal activity as an index of cortical activity and called for tighter control of test conditions (Rippon, 1993). The interpretational dilemma about ipsilateral/contralateral excitation/inhibition in this context remains unresolved. Gruzelier, who pioneered research in electrodermal asymmetry in schizophrenics, has recently reiterated the strong influence of arousal, attention and anxiety on electrodermal activity and cautioned its use as a trait marker (Gruzelier and Raine, 1994).

In view of this relatively negative literature, we propose to take a step back and begin a new approach using objective measures, detailed quantification and minimal assumptions. Our approach is to study a reliable cortical measure and its relationship with the simplest form of electrodermal activity during performance in a simple paradigm. We attempt to answer the first of a series of basic questions: what is the relationship between tonic SCL and EEG during a simple task in a condition known to be conducive to arousal decline. Despite considerable temporal (Niedermeyer, 1987a; Inouye et al., 1991) and inter-subject (Chatrian and Lairy, 1976; Niedermeyer, 1987a) variability, the EEG frequency spectrum was chosen as a CNS index because it is an historically important measure of cerebral cortical activity and because of its sensitivity to arousal fluctuation (Evans, 1993; Makeig and Inlow, 1993), its reasonable test-retest reliability (Gasser et al., 1985) and its ease of access. Specifically, this study explores the *prestimulus* EEG spectral components relating to *prestimulus* SCL during a repetitive stimulus paradigm. It is an extension of the Barry and Sokolov (1993) study which examined prestimulus SCL alone. We seek to establish a quantitative relationship between cerebral cortical and peripheral autonomic measures of arousal. Quantification of the ANS/CNS relationship may help to evaluate this confounding factor in cognitive research and may stimulate the development of methods for its compensation.

2. Methods

Ten normal right handed volunteers (6 men and 4 women) aged 21–51 years (mean = 31.7; sd = 9.0 years) were recruited for the study. They were seated comfortably in a quiet, dimly lit laboratory with air-conditioned ambient temperature set at 24 ± 1°C. Subjects sat facing a monitor screen and wearing a pair of headphones. An auditory 'habituation' paradigm consisted of 22 trains of 7 tones. The 7 tones, each of 80 dB SPL and lasting 50 ms, were presented with an inter-stimulus interval (ISI) of 10 seconds, with a change in the pitch from 1000 Hz to 1500 Hz at the 6th tone, and an inter-train interval (ITI) of 20 s. Subjects were instructed to fixate on a coloured dot on the screen and count the number of times it changed from red to blue. They were told that they could ignore the tones, which were not relevant to their task.

Electroencephalogram (EEG), electro-occulogram (EOG) and skin conductance level (SCL) were recorded simultaneously and continuously for 30 min using a 32-channel PC based system. The signals were continuously digitised at 256 Hz and stored to a hard disk. EEG was recorded from 19 head sites (Fp1,Fp2,F7,F8,F3,F4,Fz,T3,T4,C3,C4,Cz,T5,T6, P3,P4,Pz,O1 and O2) placed according to the 10–20 system (Jasper, 1958) in reference to linked-ear electrodes, with an amplification of 50,000 and a bandpass from 0.07 Hz to 50 Hz. A pair of electrodes was placed above and below the eye to record vertical EOG and two electrodes were placed at the outer canthus of each eye to record horizontal EOG. The bandpass was the same as for the EEG channels. Skin conductance was recorded via a pair of silver-silver chloride electrodes filled with 0.05M sodium

chloride gel placed on the distal phalanges of digits II and III of the non-dominant hand after the skin area was wiped with an alcohol swab. The electrode pair forming part of the input circuit was excited by a constant voltage and hence the current change representing conductance was recorded using a DC amplifier. EOG correction was performed using a technique based on Gratton et al. (1983). After EOG correction, any prestimulus EEG epoch containing signal in excess of ± 500 μV, and the corresponding SCL, was excluded from analysis.

Fast Fourier transforms (FFT) of 154 EEG samples of the 2-s epoch *prior* to each tone were then performed after applying a windowing function to minimise the leakage and end effects. The EEG power spectra before each tone were grouped into 9 band powers (in μV^2) from 0.5 Hz to 30 Hz as follows: – two delta bands, $\delta 1$ and $\delta 2$ (0.5–1.5 Hz and 2.0–4.0 Hz), theta band (4.5–7.5 Hz), three alpha bands, $\alpha 1$, $\alpha 2$ and $\alpha 3$ (8.0–9.5 Hz, 10–11.5 Hz and 12–13.5 Hz) and three beta bands, $\beta 1$, $\beta 2$ and $\beta 3$ (14.0–18 Hz, 18.5–24 Hz and 24.5–30 Hz). These data at the 154 tones are referred to as EEG 'tone data'. The EEG tone data of each train (7 tones/train) were averaged to obtain the EEG 'train data'. Mean EEG powers of each of the 9 EEG bands, of the four classical bands (delta 0.5–4.0 Hz, theta 4.5–7.5 Hz, alpha 8.0–13.5 Hz and beta 14–30 Hz), and the total power for each train across 10 subjects were also obtained. A curve as a function of train sequence was fitted by the method of least squares to each EEG band power data set in order to show the general trend of EEG variation with time.

Similarly to the EEG tone data, the SCL (in μS) for each 2-s epoch *prior* to each stimulus tone was obtained by taking the mean within the epoch, and these are referred to as the SCL tone data. The SCL tone data associated with each train were averaged to obtain the SCL train data. The mean SCL for each tone and mean SCL for each train across the 10 subjects were obtained. In order to characterise the overall variation of skin conductance in individuals and in the group, the 154 SCL tone data points were plotted in the order of tone presentation. A curve was fitted to the group SCL data as a function of tone sequence.

Pearson's correlation analysis was performed on the train data between SCL and total EEG power, combined theta, alpha and beta powers (tab) and each of four EEG band powers at the three middle head sites in the 22 train data sets. Similar correlation studies were performed between SCL and each of the 9 EEG band powers at all head sites. The topographic distributions of the correlation coefficients and also of the associated significance levels of the train data were mapped by interpolating the data into 13 false colours on a realistic head surface. The head surface was determined by a truncated spherical harmonic expansion of order 16 based on a critical set of actual measurements of a head surface (Lim et al., 1989; Grognard and Seagar, 1989). The green colour represents zero correlation coefficient and increasingly warmer colours denote larger positive correlation coefficients until red = 1.0. Increasingly cooler colours denote larger negative correlation coefficients until violet = -1.0. The colours of the significance level maps range from red for p-values equal to 0.000 through cooler colours for increasing p-values to violet for p-value = 1.000. A multiple linear regression of SCL was also performed, using 27 covariates (9 EEG band powers from Fz, Cz and Pz). The individual correlation coefficient values were z-transformed before averaging and then inverse z-tranformed.

3. Results

3.1. Skin conductance level (SCL)

SCL variation over the 154 tone presentations for a typical individual is shown in Fig. 1. Considerable deviation from this SCL pattern was seen in two subjects, but they were still included in the group data analysis. The plot of group SCL variation (Fig. 2) shows a similar pattern to the individual, with a rising SCL over the first few tones, followed by an exponential decline. Two exponential terms were needed to fit the observed data. The fitted equation in terms of tone sequence number (t) was an algebraic sum of an increasing exponential term and a decreasing exponential term with distinct initial levels (a_1 and a_2) and time constants (t_1 and t_2) in the form:

$$f(t) = a_1(1 - \exp(-t/t_1)) + a_2(\exp(-t/t_2)); \quad t_2 \gg t_1; \qquad (1)$$

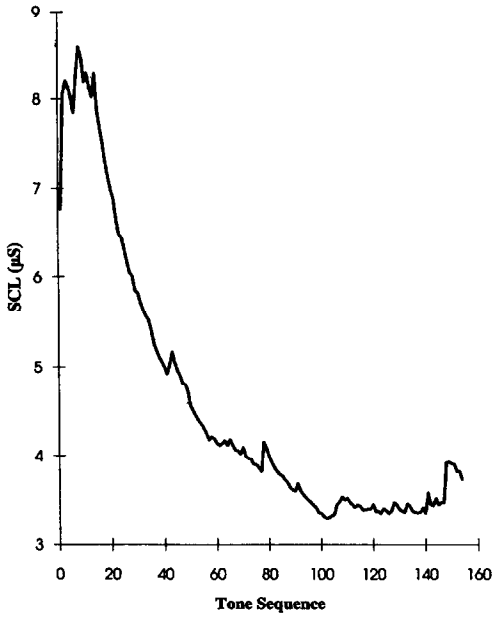


Fig. 1. Variation of SCL over the entire tone sequence in a typical individual subject. Note the initial increment of SCL followed by a prolonged exponential decline.

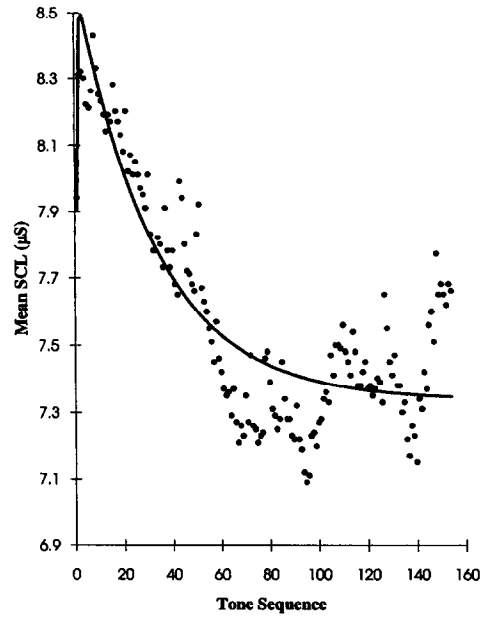


Fig. 2. Plot of SCL over the entire sequence averaged across subjects. Note the pattern of variation similar to that of the individual case shown in Fig. 1. There was a large rise during the first few tones and a steady decline from the 8th tone to the 70th tone and thereafter a general increase in variability. The solid line is the fitted curve (Eq. 1) consisting of the sum of an excitation (sensitization) and an exponential decline (habituation).

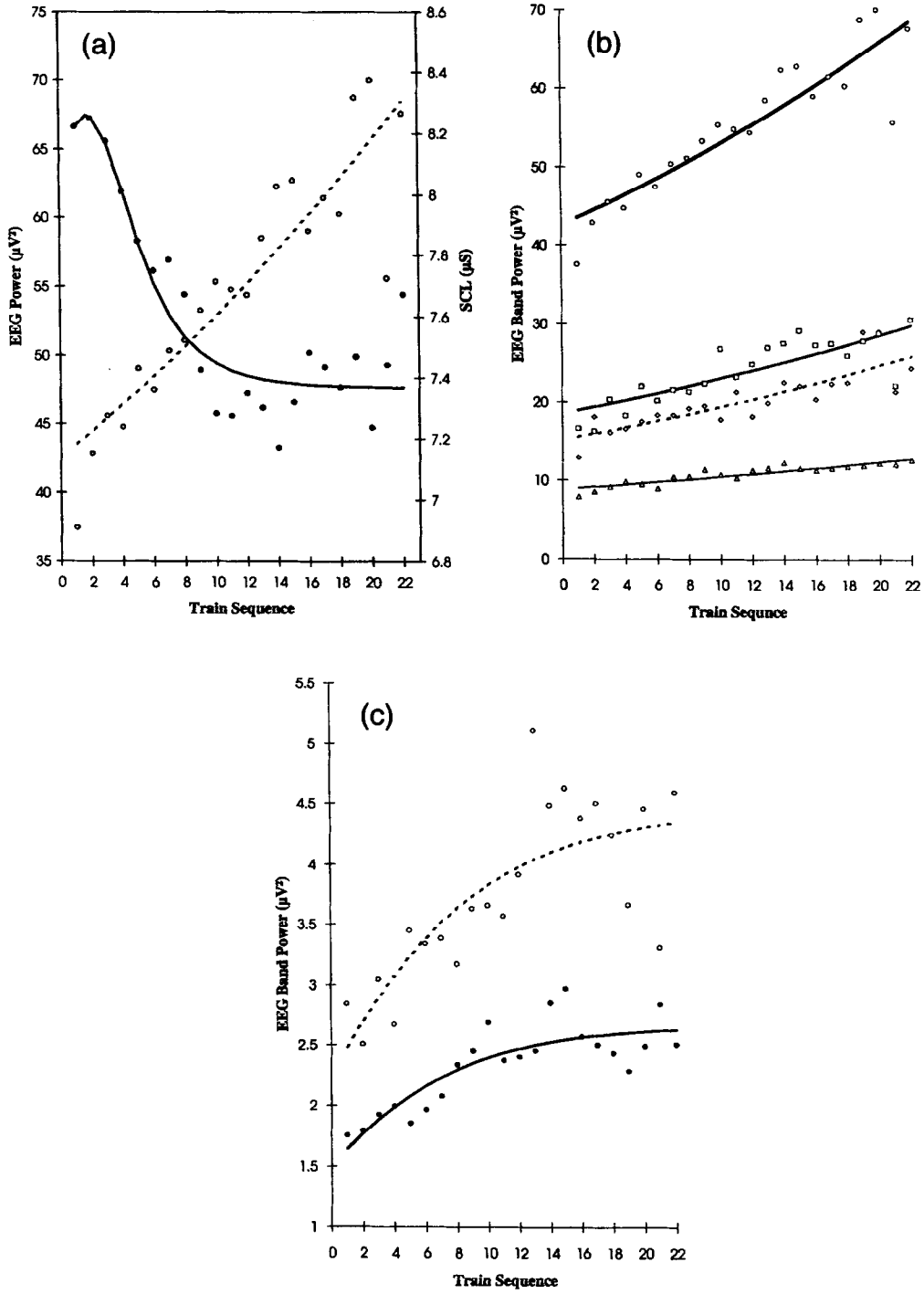
where $a_1 = 7.337$, $a_2 = 1.279$, $t_1 = 0.4182$, and $t_2 = 31.75$

Solving the equation for $df(t)/dt = 0$, we obtained the maximum SCL at $t = 1.7$, or approximately at tone 2.

3.2. Quantified EEG

None of the EOG-corrected EEG epochs contained excessively large amplitude and hence all the prestimulus EEG epochs were accepted for spectral analysis. Despite considerable inter-subject variations in EEG band powers, the mean EEG powers averaged across 10 subjects in the conventional theta, alpha, beta bands and the total EEG for each train exhibit a generally increasing trend with train sequence. For example, the variations with train sequence of the theta, alpha and beta band and their combined band power at Cz are shown in Fig. 3b. This trend is apparent also in a large number of fine-grain EEG band powers, for example, $\beta 3$ power at Fz and $\alpha 2$ power at Cz (Fig. 3c). All these trends were opposite to the SCL trend and, for example, the opposing variations with train sequence of the combined theta, alpha and beta band power and SCL are clearly seen in Fig. 3a. The delta band power did not show any trend over the entire test period.

Fig. 3. (a) Plot of the sums of conventional theta, alpha and beta band powers (○) and the fitted curve (- - -), the SCL (●) and the fitted curve (————) against the train sequence. Note the opposite trends of EEG power and SCL variations during the test period. (b) Plot of conventional beta (△), alpha (□), theta (◇) band powers at Cz and their sum (○) together with their fitted curves against the train sequence. Note the generally increasing trends. (c) Plot of mean $\beta 3$ band power at Fz (●), its fitted curve (————), mean $\alpha 2$ band power at Cz (○) and its fitted curve (- - -) against the train sequence. Note that both the band powers generally increase with the train sequence up to approximately 15 min and then ease off to their plateaux. The vertical scale readings should be multiplied by a factor of 2 for $\alpha 2$ power.


(a)
EEG Power (μV²)
SCL (μS)
Train Sequence
(b)
EEG Band Power (μV²)
Train Sequence
(c)
EEG Band Power (μV²)
Train Sequence

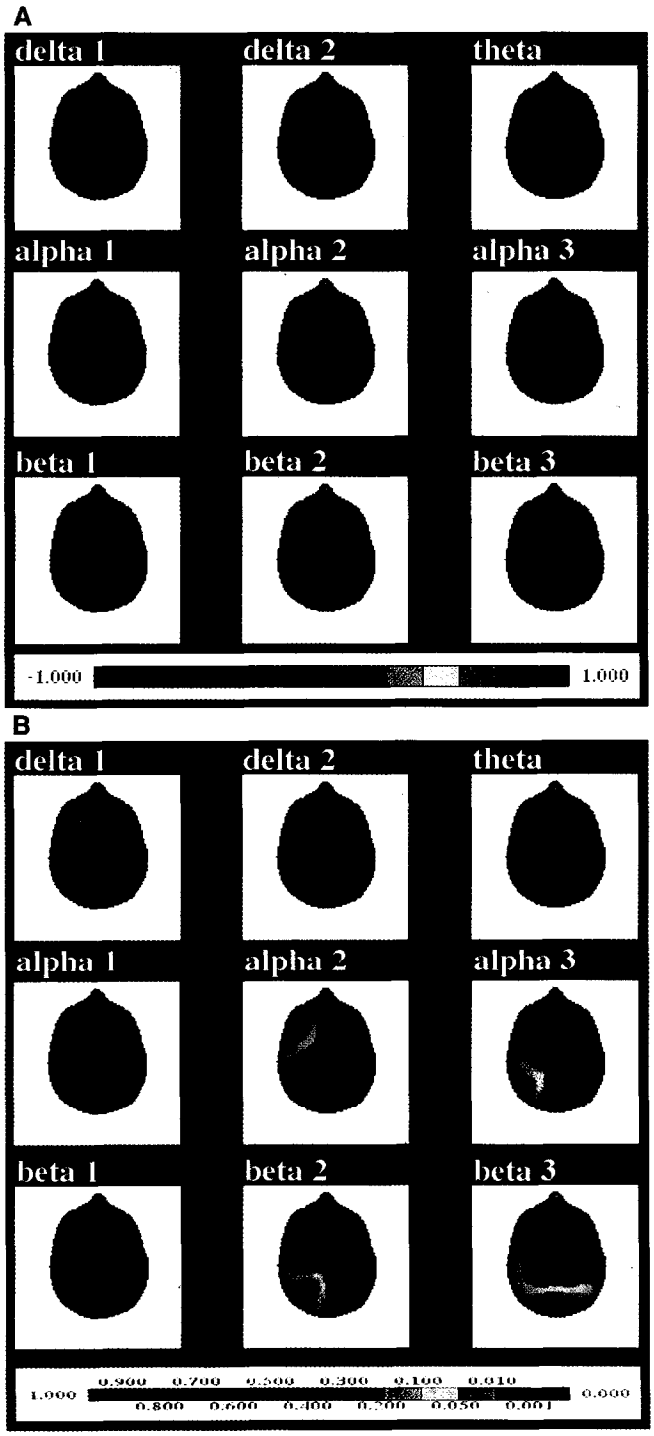

A
delta 1
delta 2
theta
alpha 1
alpha 2
alpha 3
beta 1
beta 2
beta 3
-1.000
1.000
B
delta 1
delta 2
theta
alpha 1
alpha 2
alpha 3
beta 1
beta 2
beta 3
1.000
0.900 0.700 0.500 0.300 0.100 0.010
0.800 0.600 0.400 0.200 0.050 0.001
0.000

Table 1
Regression data between SCL and EEG power

EEG variable	Regr. coeff.	p-value	Corr. coeff.
EEG Fz	−0.021	0.00113	−0.647
EEG Cz	−0.026	0.00006	−0.748
EEG Pz	−0.017	0.00096	−0.654
Beta Fz	−0.200	0.00001	−0.797
Beta Cz	−0.208	0.00000	−0.813
Beta Pz	−0.283	0.00012	−0.728
Alpha Fz	−0.075	0.00039	−0.689
Alpha Cz	−0.063	0.00001	−0.789
Alpha Pz	−0.027	0.00164	−0.631
Theta Fz	−0.033	0.01470	−0.513
Theta Cz	−0.054	0.00217	−0.618
Theta Pz	−0.046	0.00557	−0.570
Delta Fz	−0.014	0.37486	−0.199
Delta Cz	−0.006	0.71607	−0.082
Delta Pz	−0.009	0.64256	−0.105

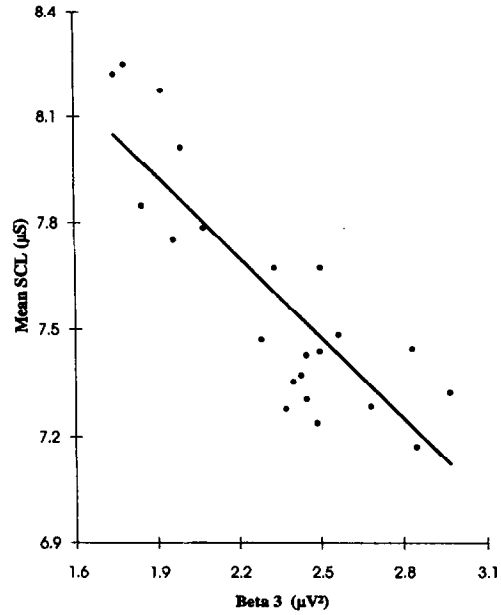


Fig. 5. Plot of SCL (μS) against $\beta 3$ band power (μV^2) at Fz, together with the best fitted linear regression: SCL = −0.756 $EEG_{\beta 3_Fz}$ + 9.37; $r = -0.82$, $p < 0.00003$.

3.3. SCL and EEG correlation

Correlation between theta, alpha and beta band powers and SCL in the group data was stronger than within the data of any individual. The EEG band powers and SCL in the train data exhibited higher correlation coefficients than those of the tone data. The correlation study was therefore performed mainly on the train data. The regression coefficients, p-values and correlation coefficients between group mean SCL and the four traditional EEG band powers at the midline sites are shown in Table 1. Note that all but the delta band powers are significantly negatively related to SCL. The more detailed results of Pearson's correlation analysis for the train data for each of the 9 EEG band powers at each site are more concisely shown using two series of topographic maps. A series of topographic maps of correlation coefficients and a series of significance levels are shown in Fig. 4. SCL showed a significant negative correlation with beta, alpha and also, to a lesser degree, theta band powers. Beta correlates were widely distributed, with dominance in the posterior head region for $\beta 1$ and in the fronto-central head region for $\beta 2$ and 3. The strongest correlation was with $\beta 3$ at Fz (SCL = −0.756 $EEG_{\beta 3_Fz}$ + 9.37; $r = -0.82$, $p < 0.00003$) (Fig. 5). Alpha correlates were less extensive and appeared mainly over the central head region. The strongest alpha correlate was $\alpha 2$ at Cz (SCL = −0.159 $EEG_{\alpha 2_Cz}$ + 8.79; $r = -0.71$, $p < 0.0002$). SCL correlation with theta band power was not as strong as the beta and alpha bands. There were no significant correlations with delta band power. An exhaustive search procedure (Lunn and McNeil, 1991; Gelbski et al., 1992) was used to find the best fitting multi-variable model for SCL considering all the 9 EEG band powers at the 3 central sites (Fz, Cz and Pz) as possible covariates. Only $\beta 3$ band power at Fz ($EEG_{\beta 3_Fz}$) and $\alpha 1$ band power at Fz ($EEG_{\alpha 1_Fz}$) were required in the final model for the SCL in the train data across subjects. The multi-variable model obtained for SCL was: SCL = −0.685 $EEG_{\beta 3_Fz}$ − 0.045 $EEG_{\alpha 1_Fz}$ + 9.556; $r = -0.89$.

Fig. 4. A: Topographic display of coefficients of Pearson's correlation between SCL and each of the 9 EEG band powers, based on group train data. These maps should be interpreted with the significance level maps shown in B. Note the widespread *negative* correlation coefficients for beta, alpha and theta band powers. There were strong SCL correlations with the beta and the alpha band powers. B: Significance level maps in which the three warmest colours indicate $p < 0.05$. These maps show that SCL was not correlated with the delta band powers, was correlated weakly with theta power, and strongly with the alpha band and beta band powers.

The significance levels for the three coefficients being non-zero were respectively $p < 0.0001$, $p < 0.03$ and $p < 0.0001$. The F statistic was 31.86 and the probability of the two variables as covariates was less than 0.00001.

4. Discussion

Concurrent studies on prestimulus background SCL and EEG are rare, and their quantitative relationship, to the best of our knowledge, has not been reported. We shall discuss our findings for these measures in turn, and then their relationship, bearing in mind that the quantitative results were obtained from a small number of subjects. While they provide strong evidence to suggest useful trends, their generalisation should be guarded.

4.1. SCL

The group prestimulus SCL in subjects exposed to a series of brief indifferent auditory stimuli showed a large initial rise, followed by a systematic slower exponential decrement with time, consistent with the profile of initial increment and exponential decline reported by Barry and Sokolov (1993). The general decline of SCL with time has been reported in other studies (Davies and Krkovic, 1964; Bohlin, 1971; Gruzelier, 1973; Siddle et al., 1974; Elie and Guiheneuc, 1990), but those earlier experiments were not designed to detect the initial transient changes. Our finding that 2 of our 10 subjects did not demonstrate this SCL profile is consistent with the finding of Schell et al. (1988). Variable changes have also been reported in a long recording period of 2 h (Ross et al., 1959). In our study there was an increased SCL variability about an asymptote after approximately 15 min of recording, which continued up to the end of the 30 min recording session. The variation up to 15 min in this study is similar to the figure of 12–13 min noted by Bohlin (1971) and also by Siddle et al. (1974). The larger variability about an asymptote, suggesting increased fluctuating electrodermal activity, probably is caused by a marked reduction of tonic arousal, when nonspecific SCL responses become dominant, perhaps in response to internal activation (Beyer et al., 1990; Nikula, 1991).

The SCL changes may be described by Eq. (1) as a function of time. The first exponential term of short time constant may represent the process of sensitization (Groves and Thompson, 1970), and the second exponential term of longer time constant may represent the slower recovery of this sensitization superimposed on a tonic level reflecting falling basal arousal. The initial sensitization increase in arousal was noted by Barry and Sokolov (1993) to be compatible with Sokolov's notion of the tonic OR, but had been overlooked in earlier work. This acknowledgement provides an important theoretical link between Groves and Thompson's dual-process theory (Groves and Thompson, 1970) and Sokolov's stimulus-comparator theory of habituation (Sokolov, 1963). Both recognise arousal level as the amplifying factor in phasic response determination, and both appear to be equally efficacious in handling habituation of both phasic and tonic changes elicited by innocuous stimuli in indifferent/ignore situations. In this context, the present SCL data paint a coherent picture of graded arousal/activation changing systematically with stimulus repetition. This has substantial potential in unravelling stimulus-response relationships because of its acknowledged role in the substrate of phasic responsivity via its amplifying role.

This graded activation might be a result of a simple gain change of a complex neuronal interaction reflecting the process of tonic habituation. The gain may range from zero to a maximum, depending on variables such as stimulus novelty, past exposure and expectation. It is probably mediated by cholinergic projection from the basal forebrain to the cerebral cortex, and possibly by projection involving other neurotransmitters (Steriade et al., 1990), corresponding to phasic activation mediated by the upper portions of the activating system proposed by Sharpless and Jasper (1956). Such a model would be consistent with both the arousal system II hypothesised by Routtenberg (1968) and supported by Montplaisir (1975), and the behavioral activation system of Fowles (1980). These commonalities may provide the beginnings of a useful theoretical fusion.

The slow decrement possibly reflects a mixture of the tonic arousal level and the passive clearance of the excitation substrates, such as sweat and fluid volume, in the recovery process. The general decline of tonic SCL, similar to the decreasing frequency of

non-specific GSR with an increasing dose of intravenous pentothal (Burch and Greiner, 1960), might reflect a decreasing tonic excitation gain due to a lowering level of arousal (Parasuraman, 1983).

4.2. EEG

The systematic trend of increasing power in the conventional beta, alpha and theta bands and their sum (Fig. 3b) and specially of the fine-grain $\alpha 2$ and $\beta 3$ band powers with increasing time (Fig. 3c) can be clearly seen. The temporal variation pattern of the fine-grain EEG band powers, although in the opposite direction, appeared to be closer to that of SCL. Note that the initial fast increases, reaching their respective plateaux soon after 15 min and remaining there up to the end of the 30-min test period is similar to the SCL temporal variation. The systematic variations suggest the presence of a global factor. The most likely factor in our experimental setting is a declining arousal level with time. Increasing EEG power with decreasing arousal levels had been reported in cats, monkeys and humans. Using implanted electrodes in cat amygdala and cerebral cortex, Pagano et al. (1963) showed a systematic increase of cortical EEG activity and of amygdala fast activity with decreasing levels of behavioural arousal. Short phasic EEG desynchronisation was seen in cortical EEG recorded from chronically implanted microelectrodes in monkey even after the tonic cholinergic pathway was blocked by hyoscine (Montplaisir, 1975), a finding consistent with the tonic and phasic dual cortical activation systems proposed by Sharpless and Jasper (1956). The firing rate of some neurones at the locus coeruleus was shown to increase with decrement of EEG power between 4 and 20 Hz when the monkey's arousal level was mildly raised (Swick et al., 1994). The time course of local error rates in performing a target tone detection task, as reported by Makeig and Inlow (1993), paralleled changes in the EEG spectrum and is thought to be an expression of cyclic variation of arousal. Note the similarity between the temporal profile of the EEG spectrum over a period of 28 min (their Fig. 3) and our $\alpha 2$ and $\beta 3$ variations over a similar duration.

The temporal variation of EEG band power is consistent with a behavioural arousal continuum. Fruhstorfer and Bergstrom (1969) reported difficulty in keeping subjects awake after 10–15 min of a vigilance study. The decline to drowsiness during unattended repetitive stimulation has been reported as both common and rapid, occurring in a matter of 3 (Tizard, 1966) to 10 min (Siddle et al., 1974). Bohlin (1971) reported sleep onset after 20 min during monotonous stimulation. All these findings suggest that the systematic change of EEG powers in our series reflects declining arousal and most probably represents the normal temporal pattern of arousal variation in a simple laboratory situation.

4.3. SCL and EEG relationships

The EEG train data correlated better with the SCL than did the EEG tone data, probably because the train data decreased the effect of EEG variability over a short time scale. The changes in beta, alpha and theta EEG band powers occurred in a direction opposite to SCL (Fig. 3a and Table 1). The total EEG power was significantly related to SCL even though the delta band power was not related to SCL.

The distribution of the correlation coefficients of beta activity with SCL was consistent with the topographical distribution of these rhythms (Chatrian and Lairy, 1976). The SCL correlated with the lower frequency (14–16 Hz) $\beta 1$ mainly over the posterior head region, whereas $\beta 3$ had high levels of correlation in the fronto-central region. On the other hand, $\beta 2$, which is considered to be diffuse and nonspecific, appeared to change in parallel with the other two bands and had a distribution mainly in the anterior and central head regions. The consistency of our beta significance level maps with the beta distribution reported in the neurophysiological literature indicates that the beta activities and SCL are mediated diffusely by common neuronal networks of the reticular-thalamo-hypothalamo-cortical tonic activating system. Belyavin and Wright (1987) found beta activity a good discriminator for worsening vigilance level. In a study of 26 normal adults Rockstroh et al. (1988) reported increased electrodermal activity in association with the more aroused contralateral hemisphere. These findings, including ours, bear testimony to the fact that the cerebral cortex and the autonomic circuits are structurally linked.

The low alpha and beta activity in our initial test period is consistent with the joint decrement of alpha and beta activity during infrequent auditory stimula-

tion and during performance of more complex tasks (Van Dijk et al., 1992) and also with enhanced electrodermal activity on the hand contralateral to the cerebral hemisphere showing reduction of alpha and beta activity during verbal and visuospatial tasks (Rippon, 1990). The increment in beta activity associated with decreasing arousal seen in the second half of our study parallels the increase with drowsiness reported by Matousek and Petersén (1983). It is also in keeping with the greater suppression of beta and also alpha band powers in the left hemisphere during verbal and spatial cognitive tasks (Davidson et al., 1990; Sobotka et al., 1992), but at variance with Volavka et al. (1967), who found an increase in beta activity with cognitive activation.

There seems little doubt that EEG desynchronisation is associated with increased cognitive demand (Dujardin et al., 1993b), voluntary movement (Derambure et al., 1993), behavioral arousal (Weinberger and Lindsley, 1964) and cortical arousal (Jasper, 1936; Moruzzi and Magoun, 1949; Sharpless and Jasper, 1956; Glickman and Feldman, 1961). However, there has been a tendency to equate EEG desynchronisation with increased beta activity (Cobb, 1963). Desynchronisation or low voltage fast activity is considered to be due mainly to reduced synchronisation in many smaller aggregates of neurones with the random phase of local alpha rhythms (Lindsley, 1952; Lopes Da Silva et al., 1973) and may not necessarily be associated with an increment in beta power. Reports of the relationship between beta activity, cortical activation and behavioral arousal are rare. Some beta activity, as recorded from the scalp, probably represents fast activity in various circumscribed neocortical areas responding to focussed attention (Steriade et al., 1990) or increased arousal (Cole, 1989). The apparent contradictory findings may merely reflect the complexity and the heterogeneity of beta activities (Pantev et al., 1991) coupled with differences in experimental design, resulting in multiple interactions between beta activity, arousal, attention and activation tasks. Alpha rhythm is attenuated with alertness (Dujardin et al., 1993a; Gruzelier et al., 1993), is enhanced in a relaxed wakeful state and disappears with increasing drowsiness (Niedermeyer, 1987a) with eyes closed (Stennett, 1956). With eyes open, the appearance of alpha with visual inattention has been reported (Gibbs and Gibbs, 1950). Our observed concomitant increment of the frontocentral beta and the central alpha activities, in association with decreasing arousal, as indicated by SCL decrement, suggests that the subjects were probably in a declining state of arousal, perhaps even towards the drowsy state described as 'transitional' by Santamaria and Chiappa (1987). They found that one third of their 55 subjects had frontocentral 25–30 Hz beta in the transitional and post-transitional stage. Prevalence of beta activity of 15–25 Hz in the state of light drowsiness is widely recognised (Niedermeyer, 1987b; Kellaway, 1990).

At a practical and immediate level our findings indicate that cerebral state needs to be taken into account as a variable in any neuropsychophysiological study involving a test duration longer than a few minutes. The common functional link and the quantitative relationship noted here between central and autonomic measures may lead to the development of tools to fractionate the cortical phasic response from tonic levels, taking advantage of the separate peripheral arousal measure. The use of a separate measure for a compensation of arousal variation is made easier and more attractive in view of the linear relationships between the SCL and some fine-grain frequency components (Table 1 and Fig. 5).

In summary, the time courses of EEG activity and SCL were similar in response to a simple laboratory task conducive to physiological arousal decline. These common features are an expression of their close structural and functional links. Our findings indicate that a declining tonic level of arousal may be detected by both EEG and SCL and that there is a quantitative link between these measures of cerebral function and autonomic arousal. Further exploration of the implications of these results in relation to our understanding of the basal substrates of phasic responsivity is clearly warranted. In particular, better measures of cortical/peripheral arousal should illuminate the problem of the basis of differential responding in the clinical context for a number of neuropsychiatric disorders.

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